SAI YASASWINI MAJETY

Senior Product Software Engineer | Python Full Stack & Cloud Systems Architect

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Professional Summary

- **Over 8+ Years of Production Software Engineering Experience:** Senior Full Stack Developer specializing in high-concurrency Python backend microservices, modern React frontend architectures, and enterprise cloud & AI platforms across Intuit (Contractor) and Viasat (Full-Time).
- **Enterprise AI & RAG Microservices (Intuit):** Architected and deployed scalable AI-powered backend applications using Python 3.12, FastAPI, FAISS vector indexing, and transformer-based embeddings, delivering sub-200ms semantic retrieval and real-time React operational evaluation dashboards.
- **Python Backend & Microservice Architecture:** Specialized in asynchronous Python (FastAPI, asyncio, uvloop, Pydantic v2, Django, Flask, Celery) developing high-throughput RESTful APIs, OpenAPI contracts, and microservices supporting 12,000+ requests/second.
- **Mission-Critical Aviation Telemetry (Viasat Inc.):** 6+ continuous full-time years sustaining a strict **99.999% availability SLA** across 1,000+ commercial aircraft for Delta Air Lines, processing continuous avionics, RF link, and edge sensor telemetry streams.
- **Relational, Spatial & Time-Series Databases:** Production engineering across PostgreSQL, PostGIS, MySQL 8.0, and TimescaleDB; skilled in spatial indexing (GiST, R-Tree), coordinate transformations, spatial geofencing joins, and query plan optimization.
- **Zero-Downtime Database Engineering:** Proven track record executing online schema migrations on multi-million row operational databases using trigger-based Change Data Capture (CDC) and shadow tables, executing atomic table cutovers in under **50 milliseconds**.
- **Edge Store-and-Forward Buffering:** Engineered asynchronous edge queueing daemons that prevented telemetry loss during satellite and cellular connection blackouts while reducing airborne satellite uplink bandwidth consumption by **65%** via Zstandard compression.
- **Enterprise Systems Integration:** Practical experience interfacing IoT sensor streams with enterprise asset management (EAM) platforms (IBM Maximo, SAP PM) to automate asset health tracking and preventative maintenance work order creation.
- **Full-Stack Engineering & Modern Web Architecture:** Hands-on expertise building responsive single-page web applications and operational dashboards in React.js, Next.js, and TypeScript with real-time WebSockets and state management.
- **Enterprise Cloud Infrastructure & IaC Automation:** Extensive production experience deploying and scaling microservices across AWS (EC2, S3, RDS, Lambda, ECS) and Azure (AKS) using Terraform, Docker, and Kubernetes with zero-downtime rolling updates.
- **API Security, Identity & Governance Standards:** Deep expertise enforcing enterprise authentication standards including OAuth 2.0, SAML 2.0, Microsoft Entra ID, and mTLS zero-trust communication across distributed services.
- **Executive Recognition & Intellectual Horsepower:** Awarded **2 CEO Performance Bonuses** directly by Viasat Chairman & CEO Mark Dankberg, 2 merit promotions, and co-inventor of published **Indian Patent 202541026299**.

Flagship Honors, Awards & Technical Credentials

- **Co-Inventor, Indian Patent Published (App. No. 202541026299):** *Modular Deep Learning Architecture for Cross-Domain Transfer and Incremental Learning*, cutting neural network retraining compute by **80%**.
- **Viasat CEO Performance Awards (2x Recipient):** Commended directly by CEO Mark Dankberg for outstanding operational stability and telemetry delivery on the Delta Air Lines fleet.
- **Viasat Spotlight Recognition Award:** Awarded for enterprise technical innovation in automating fleet-wide PKI certificate rotation and zero-touch cryptographic key deployment.
- **1st Place Winner, Viasat IoT Telemetry Hackathon:** Architected and deployed an end-to-end connected IoT sensor prototype with real-time web visualization in a 24-hour sprint.
- **Industrial Automation & Systems Training (HCL Technologies):** Enterprise IT infrastructure, automation hardware, telemetry communication protocols, and industrial monitoring.
- **Microcontroller & Embedded Systems Certification (ECIL):** Embedded C programming, telemetry hardware interfacing, PIC microcontroller architecture, and serial communication.
- **Dean's Merit Scholarship Recipient (SASTRA University):** Awarded for sustained academic excellence, ranking in the top 3% of the Electrical and Electronics Engineering cohort.

Technical Skills Matrix

Programming Languages: Python 3.11/3.12, JavaScript (ES6+), TypeScript, SQL, Bash/Shell Scripting, C/C++, MATLAB
Backend Frameworks: FastAPI, Pydantic v2, Django, Django REST Framework, Flask, Celery, SQLAlchemy, Alembic, Polars, uvicorn, Gunicorn
Frontend Technologies: React.js, Next.js, TypeScript, TailwindCSS, HTML5, CSS3, Mapbox GL JS, Chart.js, Vite, WebSockets, Redux
Databases & Spatial: PostgreSQL, PostGIS, MySQL 8.0, TimescaleDB, SQLite (WAL mode), Redis (ElastiCache), MongoDB, ElasticSearch
Streaming & Messaging: Apache Kafka, RabbitMQ, AWS SQS, AWS SNS, Server-Sent Events (SSE), MQTT, Event-Driven Architecture
Cloud Infrastructure: AWS (EC2, S3, RDS, Lambda, ECS, CloudWatch, IAM), Azure (AKS), Docker, Kubernetes, Terraform, Helm
AI, LLM & Retrieval Systems: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), FAISS, Vector Embeddings, Hugging Face Transformers, PyTorch, Semantic Search, Prompt Engineering
Testing & Code Quality: PyTest, Unittest, Mock, Playwright, SonarQube, Postman, Swagger/OpenAPI Spec Validation, TDD
Enterprise Integrations: IBM Maximo (EAM / OSLC API integration), SAP PM/ERP connectors, Jira, Confluence, Git, Splunk, Datadog
Architecture & Protocols: Event-Driven Microservices, Async I/O, CQRS, CDC Stream Processing, RESTful APIs, OpenAPI, mTLS, Protobuf, gRPC, WebSockets, Zstandard

Professional Experience

Client: Intuit **Remote**

Senior Full Stack Developer (Contractor) : Python Microservices, React & Enterprise AI Platforms *Sep 2024 - Present*

Environment: Python 3.12, FastAPI, Django, Flask, PyTorch, Hugging Face Transformers, LLMs, RAG, FAISS, Vector Embeddings, PostgreSQL, Redis, AWS (ECS Fargate, S3, RDS, IAM), Docker, Terraform, GitHub Actions, React 18, TypeScript, REST APIs

- **Architected scalable AI-powered backend microservices** in Python 3.12 and FastAPI using Docker and AWS ECS Fargate, delivering high-throughput asynchronous REST APIs for enterprise financial document intelligence workloads.
- **Engineered enterprise RAG pipelines** combining FAISS dense vector indexing with lexical retrieval and Hugging Face transformer embeddings, enabling sub-200ms semantic search over complex financial, tax, and policy documentation.
- **Developed and optimized Large Language Model (LLM) integration workflows**, implementing dynamic prompt engineering, contextual chunking, automated PII scrubbing (financial privacy guardrails), and response verification checks.
- **Built resilient data persistence and vector caching layers** across AWS RDS PostgreSQL 16 and Redis, integrating S3 object storage for multi-tenant document indexing and sub-15ms semantic cache hits under 10,000+ daily queries.
- **Implemented semantic search and vector similarity algorithms** using FAISS and Hugging Face models, accelerating unstructured document classification and tabular record matching with sub-200ms response times.
- **Developed full stack operational evaluation dashboards** using React 18, TypeScript, and Python REST APIs, enabling engineers to inspect real-time model accuracy, token attribution heatmaps, and p99 inference latency percentiles.
- **Automated cloud infrastructure and CI/CD delivery** using Terraform IaC and GitHub Actions, establishing automated container vulnerability scanning, PyTest unit/integration suites, and zero-downtime deployment rollouts on AWS.

Viasat Inc. : Commercial Aviation Platform & Security Infrastructure **Chennai, India / Carlsbad, CA**

Senior Software Engineer - I (Full-Time | Client: Delta Air Lines Commercial Fleet) *Jan 2023 - Feb 2024*

Environment: Python 3.9/3.10, FastAPI, Django, PostgreSQL, MySQL 8.0 Multi-AZ, Apache Kafka, Redis, Docker, Kubernetes, AWS, Linux, OpenSSL, PKI

- **Sustained a five-nines (99.999%) availability SLA** across 1,000+ commercial passenger aircraft for Delta Air Lines, engineering distributed backend services in Python that processed continuous avionics, RF link, and network telemetry streams.
- **Executed zero-downtime online schema refactoring** on multi-million row operational MySQL and PostgreSQL databases via trigger-based CDC and shadow tables, completing global schema switchovers with under **50 milliseconds** of table locking.
- **Automated enterprise PKI certificate lifecycle daemon** across 1,000+ aircraft terminals, orchestrating automated X.509 renewal handshakes and cryptographic rotation that earned the prestigious **Viasat Spotlight Recognition Award**.
- **Engineered edge store-and-forward queueing daemons** in Python that buffered telemetry locally on airborne servers during satellite spot-beam handovers, eliminating telemetry loss and reducing uplink bandwidth consumption by **65%**.
- **Decoupled high-velocity telemetry ingestion** from analytical persistence layers by implementing Apache Kafka message buffers, eliminating database write bottlenecks during peak transcontinental flight banks.
- **Architected secure OAuth 2.0 and SAML federation gateways** integrating airline identity providers (Delta SkyMiles) with in-cabin network authentication services, supporting over **250,000 daily passenger authentications** and cutting partner onboarding from **3 weeks to 2 days**.
- **Pioneered real-time flight telemetry visualization services** that ingested streaming avionics data via Kafka and WebSockets, rendering live satellite signal strength and antenna tracking metrics for airline network operations centers.
- **Automated multi-region failover and disaster recovery drills** across AWS and on-premise ground station networks, reducing recovery time objective (RTO) from **4 hours to under 12 minutes**.
- **Awarded 2 CEO Performance Bonuses** directly by Viasat Chairman & CEO Mark Dankberg for exceptional operational stability during large-scale airline fleet software rollouts.

Viasat Inc. : Cloud Platform & API Microservices **Chennai, India**

Software Engineer II (Full-Time | Cloud Platform & API Microservices) *Jan 2021 - Dec 2022*

Environment: Python 3.8/3.9, FastAPI, AWS (CloudFormation, RDS, ECS, S3, IAM), Jenkins, Docker, Linux, Shell Scripting, PyTest

- **Accelerated multi-region cloud deployment velocity by 65%** (from 4 hours to 45 minutes MTTR) by authoring modular AWS CloudFormation IaC templates and declarative Jenkins CI/CD pipelines across staging and production clusters.
- **Elevated test coverage from 60% to >92%** across critical avionics, billing, and session authorization microservices by architecting automated PyTest suites, synthetic health harnesses, and network failure simulations.
- **Engineered automated cross-region replication monitoring** and health failover probes for enterprise AWS RDS MySQL instances, guaranteeing transactional recovery objectives (RPO/RTO) under disaster recovery drills.
- **Automated multi-region AWS infrastructure provisioning** using modular CloudFormation templates and Shell scripts, reducing new environment provisioning time from **3 days to under 4 hours**.
- **Implemented automated database backup verification and point-in-time recovery testing** for enterprise MySQL clusters on AWS RDS, ensuring a strict Recovery Point Objective (RPO) of <5 minutes.
- **Containerized legacy avionics support microservices** using Docker and integrated automated security vulnerability scanning with SonarQube, cutting production defect escape rates by **45%**.
- **Awarded the 2021 CEO Mark Dankberg Performance Bonus** for high-impact infrastructure delivery and zero-downtime fleet reliability.

Professional Experience (Continued)

Viasat Inc. : Distributed Systems, Search & Tooling Automation **Chennai, India**

Software Engineer I & Intern (Full-Time | Distributed Systems & Tooling) *Jan 2018 - Dec 2020*

Environment: Python 3.7/3.8, Django, Flask, PostgreSQL, MariaDB, Sphinx, Robot Framework, Docker, Jenkins, Git, Linux

- **Commercial Air PAT (Pipeline Observability & Quality Dashboard):** Engineered real-time observability platform in Python Flask and React aggregating build results and defect telemetry across **50+ Jenkins nodes**, cutting release verification latency by **40%**.
- **CandEx 2.0 Platform (Internal Talent Operations & Search Engine):** Accelerated technical resume search query response times from **3.2s to <180ms** (17x speedup) by integrating Sphinx full-text search engine with Python Django microservices across 10,000+ records.
- **Automated pre-merge regression verification pipelines** in Groovy and Robot Framework, executing 100+ automated test scenarios per release candidate and preventing breaking API regressions from reaching staging environments.
- **Engineered automated ETL data extraction pipelines** that normalized heterogeneous sensor logs into PostgreSQL, accelerating daily operational reporting generation from **4 hours to 15 minutes**.
- **Built responsive web interfaces** using React and JavaScript to display real-time terminal connection status, signal-to-noise ratios (SNR), and historical uptime trends for network operations engineers.
- **Won 1st Place out of 24 international engineering teams in the Viasat Global IoT Hackathon** for designing an end-to-end connected IoT sensor prototype with cloud telemetry ingestion and live web dashboarding in 24 hours.
- **Engineered automated test harnesses** in Python for hardware-in-the-loop (HIL) avionics testing, verifying serial RS-232 and Ethernet telemetry protocols between satellite modems and airborne terminals.
- **Standardized Git branching workflows and code review standards** across the Chennai junior engineering team, cutting peer review turnaround times by **35%** and mentoring 4 incoming engineering interns.
- **Recognized with formal Viasat Leadership Appreciation Award** and received direct conversion offer from intern to full-time SWE I.

Flagship Systems Architectures & Technical Projects

Real-Time Geospatial Critical Infrastructure Monitoring SaaS Platform **FastAPI | PostGIS | Mapbox | React**

- Architected an end-to-end cloud platform in Python (FastAPI) and PostgreSQL/PostGIS to ingest high-frequency environmental sensor telemetry (air pollutants, water quality, ground vibration) from 10,000+ spatial coordinates.
- Built an asynchronous Mapbox vector tiling engine (.pbf serialization via ST_AsMVT), achieving sub-500ms dynamic map rendering of 100,000+ infrastructure assets and real-time environmental hazard boundary polygons.
- Implemented an automated EAM integration bridge connecting sensor threshold alarms to enterprise work order systems (IBM Maximo / SAP PM), automatically generating inspection tickets when environmental safety limits are breached.

High-Throughput Semantic Vector Caching Microservice **Python 3.12 | Redis | Pydantic v2 | Docker**

- Engineered a sub-20ms semantic caching proxy in FastAPI that intercepts API queries, computes embedding hashes, and performs cosine distance matching against Redis in-memory vector indices.
- Optimized data validation models using Pydantic v2 and compiled **uvloop** event loops, maintaining <5ms latency over 50MB telemetry bursts.
- Achieved an 85% cache hit rate on repetitive telemetry queries, reducing downstream database query loads by **60%** and sustaining 12,000+ requests/second under heavy synthetic load testing.

Open-Source Enterprise RAG Assistant (rag-hf-chatbot) **Python 3.11 | Hugging Face | FAISS | Streamlit**

- Engineered an interactive document question-answering system using Hugging Face transformer embeddings and FAISS vector indexing, achieving sub-200ms semantic context retrieval over complex technical documentation.
- Architected hybrid retrieval combining dense semantic embeddings with sparse BM25 keyword matching and Reciprocal Rank Fusion (RRF), elevating search relevance from 72% to **94%**.
- Implemented conversational context memory management and structured source citation grounding, eliminating model hallucination across multi-turn user dialogues.

Edge Store-and-Forward Telemetry Daemon **Python 3.11 | SQLite WAL | Zstandard | Embedded Linux**

- Built an asynchronous local queueing engine for intermittent networks using SQLite WAL ring buffering, guaranteeing zero telemetry loss during link dropouts while cutting uplink bandwidth consumption by **65%** via Zstandard streaming compression.
- Engineered crash-resilient transactional recovery routines in Python and C++, preventing database corruption and ensuring zero-packet loss during sudden inflight power cycles.

Education & Academic Credentials

SASTRA Deemed University **Thanjavur, India**

Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering (EEE) *Aug 2014 - May 2018*

Academic Distinction: Cumulative GPA: **8.2 / 10.0** | Recipient of **Dean's Merit Scholarship** (Top 3% cohort distinction).
Core Coursework: Microprocessor & Microcontroller Architectures, Control Systems, Data Structures, OOP (Python, C++), Relational Databases & SQL, Embedded Systems.
Campus Leadership & Capstone: Lead Technical Organizer, DAKSH National Techno-Management Fest; Peer Programming Mentor; Capstone: PIC Microcontroller Sensor Telemetry System (ECIL Certified).